

/backend

Execution backend surface: `/backend []` | `/backend show` | `/backend help`

Overview

`/backend` is the TUI surface for the execution backend (spec openhands FR-008). An execution backend selects **where** a session's tool invocations actually run: `local` runs tools on the host exactly as before (the default), `docker/podman` run them inside a sandbox container, and `remote` drives a second ragent server over its REST+SSE API.

With no argument the command opens an interactive **switcher panel** that shows the active backend and the health of *every* registered backend. Each row carries the backend's id, kind, health (`ok` / `unavailable` / `unknown`), and a secret-free connection summary. The panel is keyboard-driven: Up/Down move the block cursor, Enter switches the active backend to the highlighted row, and Esc closes the panel.

The panel is a modal overlay: while it is open it swallows every keystroke and the message input field stays locked.

Syntax

```
/backend          # open the switcher panel
/backend show     # alias of /backend
/backend list     # alias of /backend
/backend local    # switch to the local host backend
/backend docker   # switch to a Docker sandbox backend
/backend podman   # switch to a Podman sandbox backend
/backend remote   # switch to a remote backend
/backend <id>     # switch to a backend by its registered id
/backend help     # usage help
```

Options / Subcommands

Form	Description
<code>/backend</code>	Open the switcher panel listing the active backend and every registered backend's health
<code>/backend show</code> / <code>/backend list</code>	Alias of <code>/backend</code>
<code>/backend <kind></code>	Switch directly to a backend kind (<code>local</code> , <code>docker</code> , <code>podman</code> , <code>remote</code>)
<code>/backend <id></code>	Switch directly to a backend by its registered id (from backends in <code>ragent.json</code>)
<code>/backend help</code>	Print the usage table (no state change)

Panel keys

Key	Action
Up / Down	Move the block cursor between rows

Key	Action
Enter	Switch the active backend to the highlighted row
Esc	Close the panel

A row whose health is not ok is dimmed and **cannot** be selected: the switch is refused and the active backend is left unchanged, so the switcher can never leave the session on a backend that cannot run a tool.

Health

Kind	ok when	unavailable when
local	always (the host is always available)	never
docker / podman	the runtime binary resolves on PATH (FR-026)	the binary is missing (the detail names it)
remote	a base URL and a key are configured (FR-030)	either is absent

Health is a snapshot taken when the panel opens; the panel does not re-probe on each frame.

Persistence

A successful switch writes `execution_backend: "<kind>"` into the loaded config file (the project-local `.ragent/ragent.json` when one was loaded, otherwise the global config). The per-turn config cache is invalidated so the next turn resolves and dispatches through the new backend.

A descriptor-free kind switch (`/backend docker`) writes the bare label. To carry an image, workspace, URL, or key, declare a full descriptor in the `backends` array (or as an inline `execution_backend` object) and select it with `/backend <id>`.

Examples

Open the switcher:

```
/backend
```

Switch to a declared Docker sandbox by id:

```
/backend docker-box
```

Return to the host:

```
/backend local
```

Output

- The panel renders a bordered modal titled `Execution backend - active: <kind>`; the active row carries an `[active]` marker.
- A switch appends an `From: /backend` message bubble and sets the status to `backend: <kind>`.

- A refusal appends a [warn] message explaining why and sets the status to backend: unavailable / backend: unknown.
- No permission prompt is raised and no LLM call is made.

Related

- `/backend help` - usage table
- `docs/howtos/sandbox-backends.md` - full guide to the switchable backends, the sandbox mechanism, and a worked Docker/Podman example
- `specs/openhands/SPEC.md` - the execution backend design (local/docker/podman/remote)
- `docs/howtos/config.md` §7.36 - the `execution_backend` and `backends` config keys